Lab No: 05

Lab Title: **while Loop in C++**

- **Learning objectives:**

The objectives of this lab are:

- To understand the working of the while loop in C++.
- To learn how to implement a do...while loop.
- To understand the concept of a nested while loop.
- To use loops for solving programming problems such as printing numbers and calculating sums.

Loop

Loops are used to repeat a block of code. Being able to have your program repeatedly execute a block of code is one of the most basic but useful tasks in programming -- many programs or websites that produce extremely complex output are really only executing a single task many times. Now, think about what this means: a loop lets you write a very simple statement to produce a significantly greater result simply by repetition.

While Loop

The most basic loop in C++ is the while loop. A while statement is like a repeating if statement. Like an If statement, if the test condition is true: the statements get executed. The difference is that after the statements have been executed, the test condition is checked again. If it is still true the statements get executed again. This cycle repeats until the test condition evaluates too false.

A **while loop** is a control flow statement that allows code to be executed repeatedly based on a given condition. The *while* consists of a block of code and a condition. The condition is first evaluated and if the condition is true the code within the block is then executed. This repeats until the condition becomes false

- Syntax

Basic syntax of while loop is as follows:

```
while (expression)
{
Single statement
        Or
Block of statements;
}
```

- The expression can be any combination of Boolean statements that are legal. Even, (while x ==5 || v == 7) which says execute the code while x equals five or while v equals 7.

➢ Example:

This program prints numbers from **1 to 5** using a while loop.

```
#include <iostream>
using namespace std;

int main()
{
   int i = 1;

   while(i <= 5)
   {
      cout << i << endl;
      i++;
   }

   return 0;
}
```

➢ Nested While Loop:

A **nested while loop** is a loop placed inside another while loop. The inner loop executes completely for every iteration of the outer loop.

Syntax

```
            while(condition1)
{
   while(condition2)
   {
      // statements
   }
}
```

➢ Example: This program prints numbers in a 3 × 3 pattern.

```
#include <iostream>
```

```
using namespace std;

int main()
{
    int i = 1;

    while(i <= 3)
    {
        int j = 1;

        while(j <= 3)
        {
            cout << j << " ";
            j++;
        }

        cout << endl;
        i++;
    }

    return 0;
}
```

- **Its format is:**
  do statement while (condition);

  Its functionality is exactly the same as the while loop, except that condition in the do-while loop is evaluated after the execution of statement instead of before, granting at least one execution of statement even if condition is never fulfill.

3.Do...While Loop

The **do...while loop** is similar to the while loop, but the loop body executes **at least once** before the condition is checked.

Syntax

```
do
{
   // statements
}
while(condition);
```

Example Program (Do...While Loop)

This program prints numbers from **1 to 5** using a do...while loop.

```
#include <iostream>
using namespace std;

int main()
{
    int i = 1;

    do
    {
        cout << i << endl;
        i++;
    }
    while(i <= 5);

    return 0;
}
```

Lab Tasks

- **Task 1: While Loop**

    1. Write a C program that uses a while loop to display numbers from 1 to 10.
    2. Display each number on a separate line.

- **Input:**

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
do
{
```

```
    cout<<i<<endl;
    i++; /* code */
} while (i<=10);
    return 0;
}
```

- **Output:**

- **Task 2: Do-While Loop**

  Repeat the first task using do... while loop.

- **Input:**

```
➢ #include <iostream>
➢ using namespace std;
➢ int main() {
➢     int i = 1;
➢ do
➢ {
➢    cout<<i<<endl;
➢    i++; /* code */
➢ } while (i<=10);
➢     return 0;
➢ }
➢
➢
```

- **Output:**

- **Task 3: Sum using While Loop**

  1. Write a C program that prompts the user to enter a positive integer.
  2. Use a while loop to calculate the sum of all numbers from 1 to the entered integer.
  3. Display the calculated sum.

- **Input:**

```
#include<iostream>
using namespace std;
int main(){
    int num, sum=0, i=1;
    cout<<"Enter positive integer: "<<endl;
    cin>>num;
```

```
    while(i<=num){
    sum+=i;
    i++;}
    cout<<"The sum is "<<sum<<endl;
    return 0;
}
```

- **Output:**

Conclusion

In this lab, we successfully implemented **while loops** and **do...while loops** in C++. The while loop was used to print numbers from 1 to 10, while the do...while loop demonstrated how a loop executes at least once before checking the condition. Additionally, we used a while loop to calculate the sum of numbers up to a user-entered integer. This lab improved our understanding of loop control structures and their practical applications in programming.